Assiut University
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Intelligent Systems

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Grading

Assessment	Grade %
Final Exam	50%
Class Work	15%
Mid-Term Exam	15%
Project	20%

Text Book

Artificial Intelligence A Modern Approach
Fourth Edition

Stuart J. Russell and Peter Norvig

Content

- Chapter 2: Intelligent Agents
  - 2.1 Agents and Environments
  - 2.2 Good Behavior: The Concept of Rationality
  - 2.3 The Nature of Environments
  - 2.4 The Structure of Agents
  - Summary

Specifying The Task Environment

- Task Environments are “problems” to which rational agents are the solutions!
- To specify a task environment, four items need to be defined:
  - **P**erformance measure (GOAL)
  - **E**nvironment
  - **A**ctuators
  - **S**ensors
- That is **PEAS** description
- Taxi Driver Example !

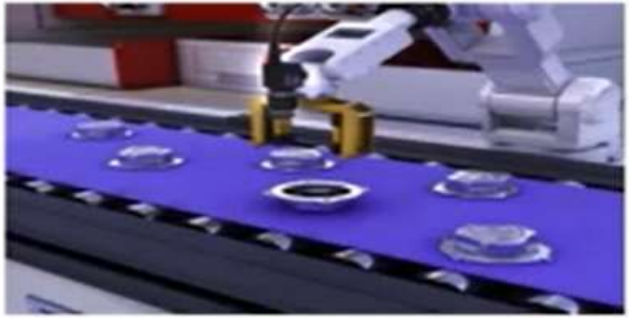
Agent Design (PEAS)

- **Performance**: How agent be assessed?
- **Environment**: What elements exists around agent?
- **Actuators**: How agent change the environment?
- **Sensors**: How agent sense the environment?

Agent Type	Performance Measure	Environment	Actuators	Sensors
Taxi driver	Safe, fast, legal, comfortable trip, maximize profits, minimize impact on other road users	Roads, other traffic, police, pedestrians, customers, weather	Steering, accelerator, brake, signal, horn, display, speech	Cameras, radar, speedometer, GPS, engine sensors, accelerometer, microphones, touchscreen

Agent Design (PEAS)

Part-picking robot



- Performance: Percentage of parts in correct bins, speed
- Environment: Conveyor belt with parts, bins
- Actuators: Jointed arm and hand
- Sensors: Camera, joint angle sensors

Agent Design (PEAS)

Medical diagnosis system

- **Performance:** Healthy patient, minimize costs, lawsuits
- **Environment:** Patient, hospital, staff,.....
- **Actuators:** Screen display (questions, tests, diagnoses, treatments, referrals)
- **Sensors:** Keyboard (entry of symptoms, patient's answers)

Agent Type	Performance Measure	Environment	Actuators	Sensors
Medical diagnosis system	Healthy patient, reduced costs	Patient, hospital, staff	Display of questions, tests, diagnoses, treatments	Touchscreen/voice entry of symptoms and findings
Satellite image analysis system	Correct categorization of objects, terrain	Orbiting satellite, downlink, weather	Display of scene categorization	High-resolution digital camera
Part-picking robot	Percentage of parts in correct bins	Conveyor belt with parts; bins	Jointed arm and hand	Camera, tactile and joint angle sensors
Refinery controller	Purity, yield, safety	Refinery, raw materials, operators	Valves, pumps, heaters, stirrers, displays	Temperature, pressure, flow, chemical sensors
Interactive English tutor	Student's score on test	Set of students, testing agency	Display of exercises, feedback, speech	Keyboard entry, voice

Examples of agent types and their PEAS descriptions.

Assignment (1)

Design the agent for your projects?
(PEAS)

Environment Properties (ODESDA)

- Observable (or, partially observable)

  An agent's sensors give it access to the complete state of the environment at each point in time

- Deterministic (or, stochastic)

  The next state of the environment is completely determined by the current state and the action executed by the agent

- Episodic (or, sequential)

  The agent's experience is divided into episodes, in each episode the agent receives a percept and then performs a single action

Fully observable vs. partially observable
Deterministic vs. nondeterministic
Episodic vs. sequential
Static vs. dynamic
Discrete vs. continuous
Single-agent vs. Multiagent

Environment Properties (ODESDA)

- Static (or, Dynamic)
  The environment is unchanged while an agent is deliberating
- Discrete (or, Continuous)
  A limited number of distinct, clearly defined percepts and actions.
- Agent (single/ multi) (cooperative /competitive)
  Number of agent in the environment

Fully observable / Partially Observable

Fully observable

- An agent can always see the entire state of an environment.
- **Example**: Chess

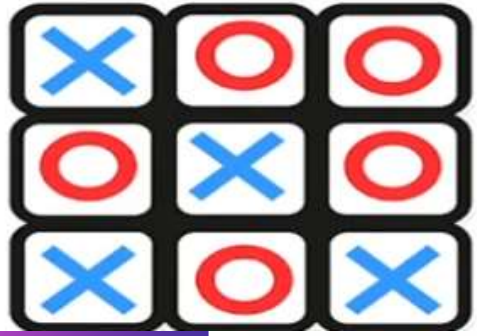
Partially Observable

- An agent can never see the entire state of an environment.
- **Example**: Card game

Deterministic / Stochastic

Deterministic

- An agent's current state and selected action can completely determine the next state of the environment.
- **Example:** Tic tac toe



Stochastic

- A stochastic environment is random in nature and cannot be determined completely by an agent.
- **Example**: Ludo (Any games that involve dice)

Episodic / Sequential

Episodic

- Only the current percept is required for the action
- Every episode is independent of each other
- **Example**: part picking robot

Sequential

- An agent requires memory of past actions to determine the next best actions.
- The current decision could affect all future decisions.
- **Example**: Chess

Discrete / Continuous

Discrete

- The environment consists of a finite number of actions that can be deliberated in the environment to obtain the output.
- **Example**: chess

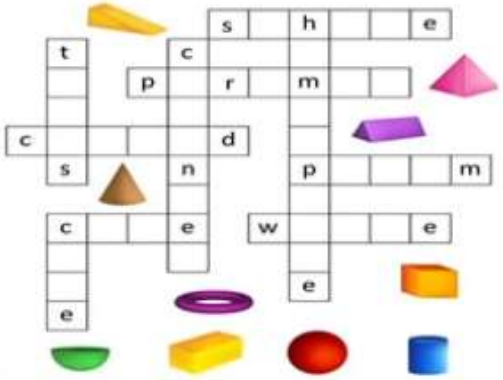
Continuous

- The environment in which the actions performed cannot be numbered ie.,is not discrete, is said to be continuous.
- **Example**: Self-driving cars

Static / Dynamic

Static

- The environment does not change while an agent is acting
- **Example**: Crossword puzzles



Dynamic

- The environment may change over time
- **Example**: roller coaster ride

  Taxi driving

Single-agent / Multi-agent

Single-agent

- If only one agent is involved in an environment, and operating by itself then such an environment is called single agent environment.
- **Example**: maze

Multi-agent

- if multiple agents are operating in an environment, then such an environment is called a multi-agent environment.
- **Example**: football

- **Fully Observable:** The agent's sensors have access to the complete state of the environment at each point in time. *Example:* A chess game — the board and all pieces are visible.
- **Partially Observable:** The agent has limited access to the full state due to noisy or limited sensors. *Example:* A robot navigating through a dark room using sonar.
- **Deterministic:** The next state of the environment is completely determined by the current state and the agent's action. *Example:* Solving a mathematical equation.
- **Stochastic:** The outcome may involve randomness or uncertainty. Example*:* Weather prediction or financial markets.
- **Episodic:** The agent's experience is divided into atomic episodes. Each episode consists of a percept and an action, and is independent of others. Example*:* Image classification where each image is processed independently.
- **Sequential:** The current decision affects future decisions. Tasks are interdependent. *Example:* Chess or autonomous driving.
- **Static:** The environment does not change while the agent is deliberating. *Example:* Crossword puzzle.
- **Dynamic:** The environment can change during decision-making. *Example:* A robotic vacuum cleaning while people walk around.
- **Discrete**: The environment has a finite number of distinct states, percepts, and actions. Example: Board games.
- **Continuous:** The environment can be described with real-valued variables. Example: Robotic arm movement, self-driving vehicles.
- **Single Agent:** The agent is the only entity acting in the environment. Example: Solving a maze.
- **Multi-Agent:** Multiple agents exist, possibly with competing or cooperating goals**.** Example: Multiplayer video games or stock trading.

Task Environment	Observable	Agents	Deterministic	Episodic	Static	Discrete
Crossword puzzle	Fully	Single	Deterministic	Sequential	Static	Discrete
Chess with a clock	Fully	Multi	Deterministic	Sequential	Semi	Discrete
Poker	Partially	Multi	Stochastic	Sequential	Static	Discrete
Backgammon	Fully	Multi	Stochastic	Sequential	Static	Discrete
Taxi driving	Partially	Multi	Stochastic	Sequential	Dynamic	Continuous
Medical diagnosis	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
Image analysis	Fully	Single	Deterministic	Episodic	Semi	Continuous
Part-picking robot	Partially	Single	Stochastic	Episodic	Dynamic	Continuous
Refinery controller	Partially	Single	Stochastic	Sequential	Dynamic	Continuous
English tutor	Partially	Multi	Stochastic	Sequential	Dynamic	Discrete

Figure 2.6 Examples of task environments and their characteristics.

Agent Basic Types:

- Simple reflex agents
- Model-based reflex agents
- Goal-based agents
- Utility-based agents

Simple reflex agents

- The simplest kind of agent.
- Choose action only based on the current percept.
- Ignoring the rest of the percept history.
- Use condition –action rules.

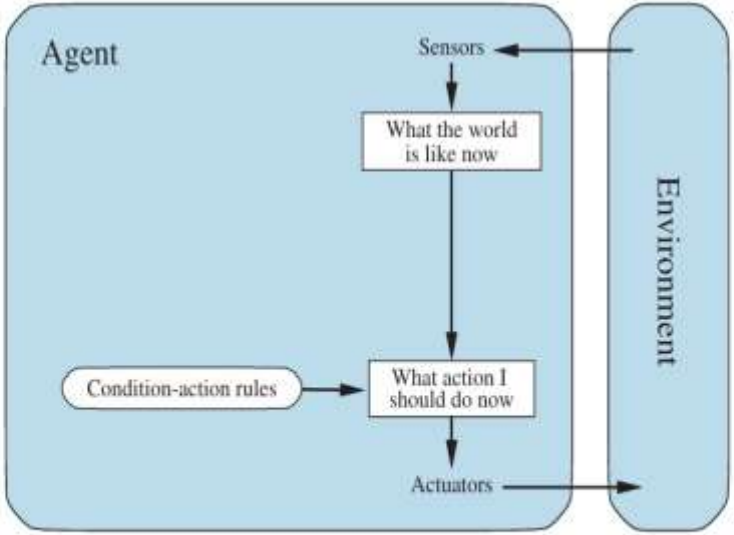


Figure 2.9 Schematic diagram of a simple reflex agent. We use rectangles to denote the current internal state of the agent's decision process, and ovals to represent the background information used in the process.

How Simple Reflex Agents Work

- Perception: The agent receives signals from the environment through its sensory devices.
- Condition Checking: It compares the input data with some pre-specified if-then rules.
- Action: It carries out the respective action using its actuators.

Key Features of Simple Reflex Agents

Simple reflex agents have basic characteristics that make them simple to comprehend yet very limited in complexity.

- **Rule-based decision-making:** It works with the "if condition → then action" principle.
- **No memory or learning:** These agents do not keep or apply any earlier decisions to future ones.
- **Quick response time:** The agent's immediate action makes it fit for the real-time task.
- **Deterministic behaviour:** The same input yields the same output consistently.
- **Limited adaptability:** Such agents cannot adapt to a new or changing environment unless they are reprogrammed.

It has no memory or internal model of the world — so it can only operate effectively in fully observable environments where every decision can be made based on the current input alone.

Example of a Simple Reflex Agent in AI

One of the best and most straightforward **simple reflex agent examples** is an automatic door sensor system.

How it works:

- **Sensors:** detect any movement going on right in front of the door.
- **Condition:** Movement is detected → then the door will be opened.
- **Action:** The door is opened automatically.

Apart from this, there are other examples such as:

- The thermostat which maintains a certain temperature in the room.
- Traffic lights that are automatically varied according to the number of vehicles present at the junction.

Smart thermostats are simple reflex agents.

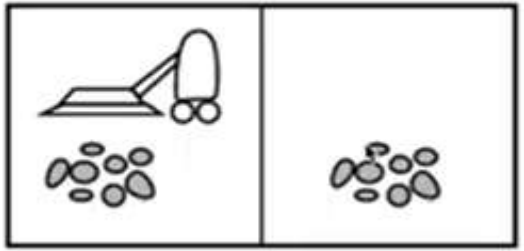
```
function SIMPLE-REFLEX-AGENT(percept) returns an action
  persistent: rules, a set of condition–action rules

  state ← INTERPRET-INPUT(percept)
  rule ← RULE-MATCH(state, rules)
  action ← rule.ACTION
  return action
```

Figure 2.10 A simple reflex agent. It acts according to a rule whose condition matches the current state, as defined by the percept.

Simple reflex agents: (dis-advantage)

- The agent will work only if the correct decision can be made on the basis of the current percept that is only if the environment is **fully observable**
- Infinite loops are often unavoidable – escape could be possible by randomizing



Model-based reflex agents

Model-based reflex agents are agents that maintain an internal memory, or "state," which helps them make decisions in situations where they cannot see or sense the entire environment directly. This memory helps them handle situations where they have only partial visibility of the environment.

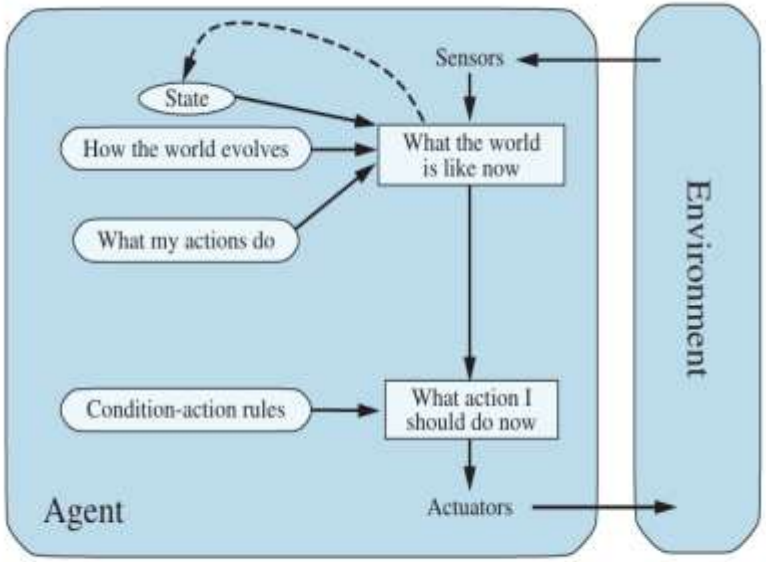


Figure 2.11 A model-based reflex agent.

How Simple Reflex Agents Work?

When it receives a new input, it updates its internal state, consults its condition-action rules, and chooses the best response based on both the current percept and what it knows from previous interactions.

❖ How does Model-based Reflex agent work ?

- Challenge: Sometimes, agents don't have a complete view of the world around them.
  - **Example:** A self-driving car agent can't see every vehicle on the road at once.
- Solution (Model-based Reflex agent):
  - The agent creates an **internal memory**, or **state**, to track things it can't directly see right now.
  - As time passes and the agent receives new information, it updates this internal state.
  - This makes its actions smarter and more responsive to changing situations.

A car agent remembering where other cars are on the road, even if they're temporarily out of sight

❖ Internal State Beyond Physical Reality

- **Beyond Just "Where" the Agent Is:**
  - The agent's memory can include goals and plans, not just information about the environment.
  - **Example:** A taxi has a rule to refuel on the way home unless it has at least half a tank.
- **Key Idea:**
  - The agent's internal state doesn't just describe the physical world around it. It also includes the agent's intentions or plans, like where it plans to go next or any steps it has planned for the journey.

A taxi with an internal rule or "goal" to refuel

Examples of Model-Based Reflex Agents

- **Self-Driving Cars:** These vehicles use sensors (cameras, LiDAR, radar) to build a 3D model of their surroundings. For instance, if a car in front is temporarily obscured by a truck, the agent's internal model remembers that the car is still there and anticipates its likely movement.

- **Robotic Vacuum Cleaners** (e.g., Roomba): Modern models create spatial maps of a home. They track which areas have already been cleaned and remember the locations of furniture to navigate more efficiently without repeating the same path.

- **Video Game NPCs**: Non-player characters use internal models to track a player's last known location. If a player hides behind a wall, the NPC "remembers" where they were and may move toward that spot rather than instantly forgetting the player's existence.

- **Smart Thermostats:** Devices like the **Google Nest** maintain a model of household schedules and local weather trends. They can start heating or cooling a home proactively before residents arrive, rather than just reacting to the current temperature.

❖ Benefits of Model-based Reflex agent work:

1. They handle **partial observability** better than simple reflex agents, as they retain context from previous observations.
2. This makes them better suited for **complex tasks**, like driving or surveillance, where the agent can't always see the entire environment.

❖ Limitations of Model-based Reflex agent work:

1. They may still struggle with environments that change unpredictably, or situations that require **extensive planning or reasoning**.
2. Additionally, creating and maintaining an accurate model can be challenging and requires **more computational power**.

Goal-based Agents

Knowing something about the current state of the environment is not always enough to decide what to do. For example, at a road junction, the taxi can turn left, turn right, or go straight on. The correct decision depends on where the taxi is trying to get to. In other words, as well as a current state description, the agent needs some sort of **goal** information that Goal describes situations that are desirable.

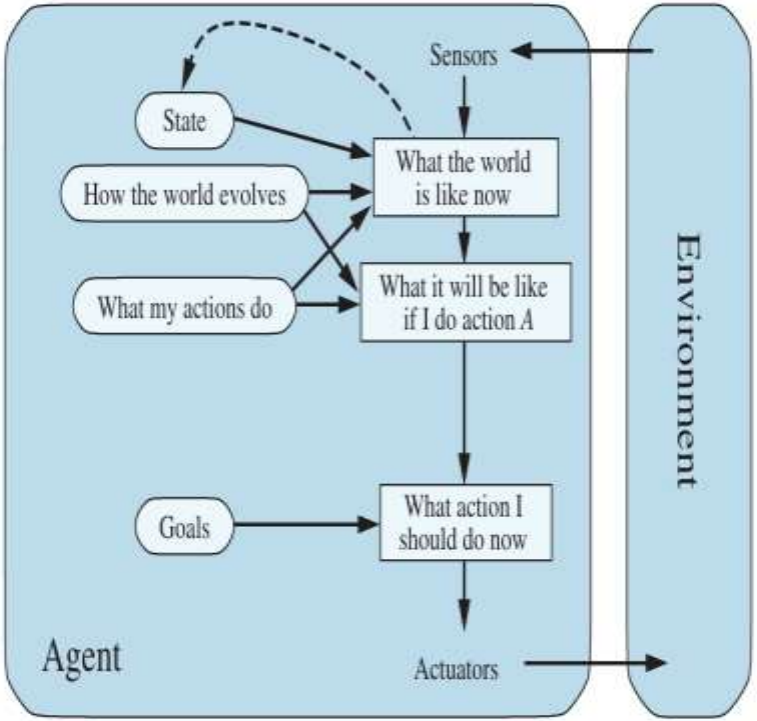


Figure 2.13 A model-based, goal-based agent. It keeps track of the world state as well as a set of goals it is trying to achieve, and chooses an action that will (eventually) lead to the achievement of its goals.

- Goal-based agents don’t just react to the current environment; they act with a goal in mind.
- Knowing only the current state isn’t enough—agents need to know where they’re going. These are an expansion of model-based reflex agents.
- Example: A taxi at a road junction doesn’t just pick a random direction; it considers its passenger’s destination to make the correct turn.

Self-Driving Car as a Goal-Based Agent

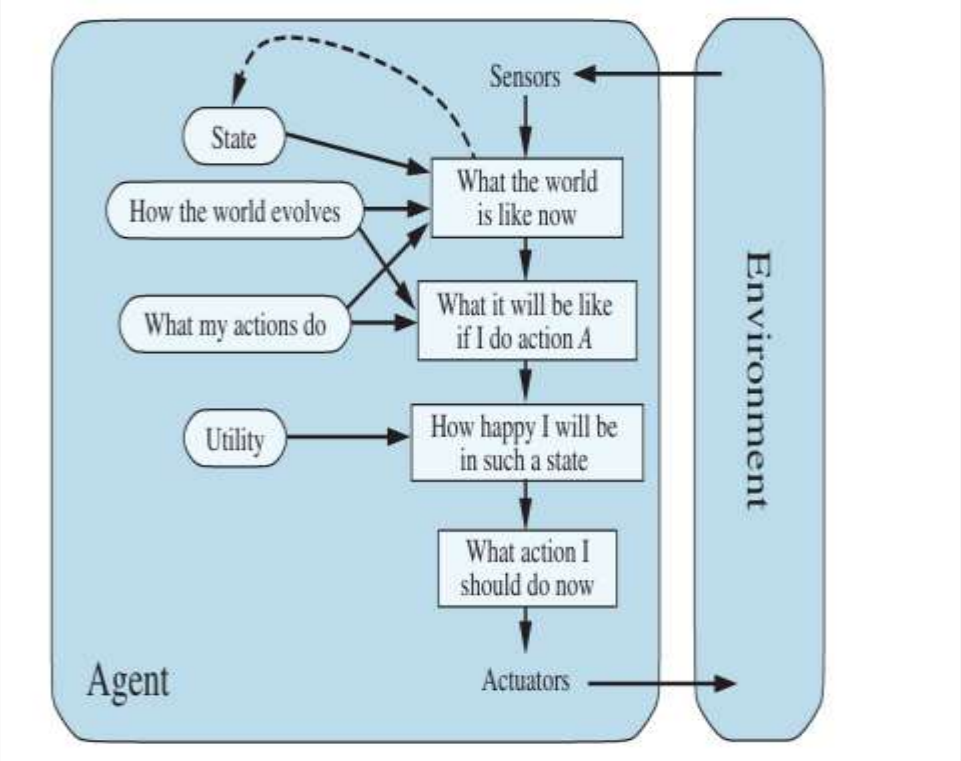
•Goal: Reach a destination safely.

•Actions are planned to avoid obstacles and follow traffic rules to reach the goal.

•Flexibility: If there’s a roadblock, the car can update its route to still achieve its goal.

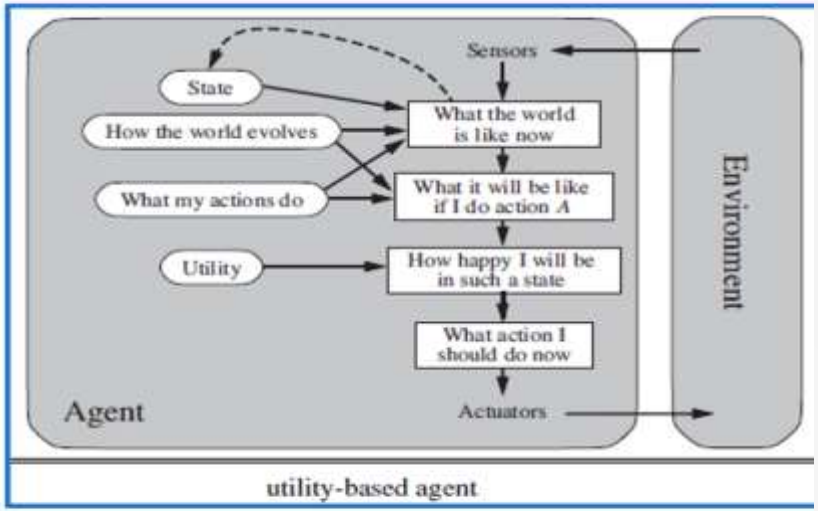
Utility-based agents

- Goals alone are not always enough for good decision-making.
- Utility-based agents go beyond "achieving a goal" and evaluate how desirable different outcomes are.
- They use a **utility function**, which measures how "happy" or "satisfied" an outcome makes them.
- Example: A taxi doesn't just aim to reach a destination—it considers whether the route is fast, safe, or cost-effective.



❖ How Do Utility-Based Agents Work?

- **Utility function:** A scoring system that ranks outcomes based on desirability.
- They calculate **expected utility**, weighing the likelihood of different outcomes against their utility.
- Decision-making involves:
  1) Modeling the environment.
  2) Predicting outcomes of actions.
  3) Selecting the action with the highest expected utility.
- **Real-Life Example:** A self-driving car picks a route balancing traffic, fuel efficiency, and safety.



- Better decision-making by balancing trade-offs (e.g., speed vs. safety).
- Flexible in handling multiple or conflicting goals.

Learning agents

What are Learning Agents ?

➢ Learning agents are intelligent systems that improve their performance over time through experience.

➢ Unlike other agents, they can adapt to new or unknown environments.

➢ They consist of key components: learning element, performance element, critic, and problem generator.

➢ Purpose: To make better decisions by learning from feedback and past experiences.

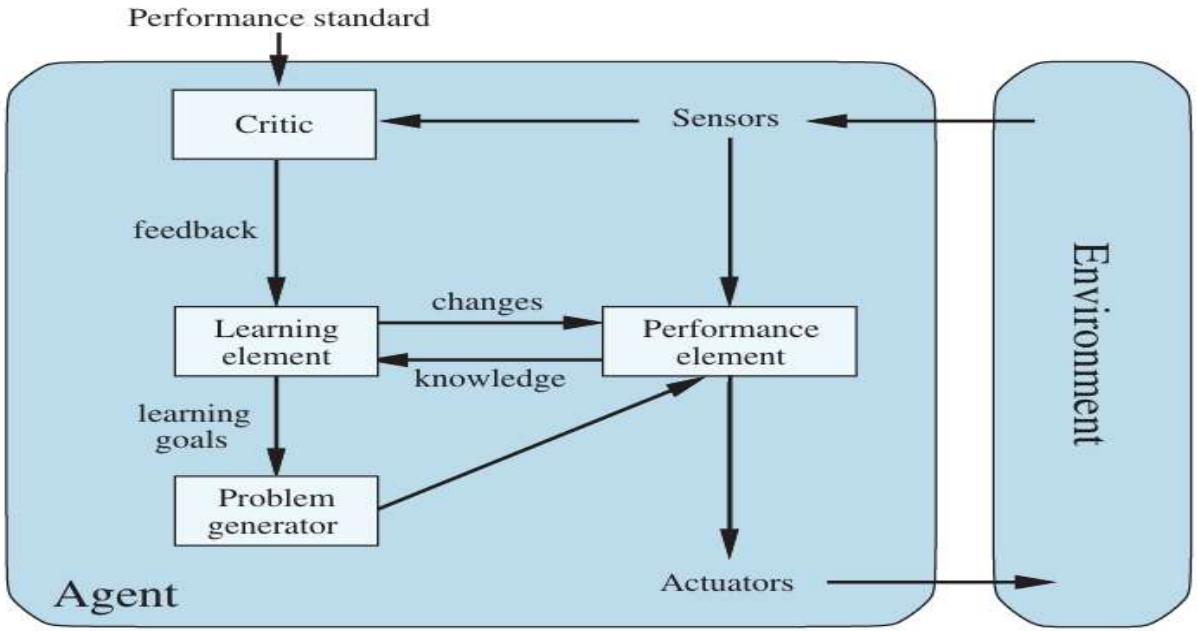


Figure 2.15 A general learning agent. The "performance element" box represents what we have previously considered to be the whole agent program. Now, the "learning element" box gets to modify that program to improve its performance.

Consider a chess playing agent, What sort of agent would it need to be?

Consider a chess playing agent, What sort of agent would it need to be?

Simple Reflex agent If yes? but some actions require some memory (e.g. castling in chess)

Model based agent If yes? but needs to reason about future

Goal based agent If yes? but what about conflicting goals?

Utility based agent Might consider multiple goals

Assignment (3)

Describe the agent type for your project?

Simple-reflex agent? Why? Why not?
Model-based agent? Why? Why not?
Goal-based agent? Why? Why not?
utility-based agent? Why? Why not?

جزاكم الله خيرا

وصل اللهم على سيدنا محمد وعلى اله وصحبه وسلم